

MTH202 QUIZ 01 SOLVED BY SAHIRULLAH

Discrete Mathematics - Complete Revision Guide

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Q1: Vertical line test is used to determine that whether the graph of a relation is a function or not.

- 1. False
- 2. True ✓

Reason: The vertical line test checks if any vertical line cuts the graph more than once. If yes, it's not a function.

Q2: In the matrix representation of an irreflexive relation all the entries in the main diagonal are ...

- 1. 2
- 2. 3
- 3. 0 ✓
- 4. 1

Reason: Irreflexive means no element relates to itself, so diagonal entries are 0.

Q3: Let $A = \{1,2,3,4\}$ and $R = \{(1,1), (1,3), (2,4), (3,1), (4,2)\}$. Which statement about R is true?

- 1. R is not symmetric
- 2. R is transitive
- 3. R is symmetric ✓
- 4. R is reflexive

Reason: Every (a,b) in R has a corresponding (b,a) . It's not reflexive because $(2,2), (3,3), (4,4)$ are missing.

Q4: Let $A = \{1,2,3,4\}$ and $R = \{(1,3), (2,2), (2,4), (3,1), (4,2)\}$. Then R is:

- 1. Symmetric ✓
- 2. Antisymmetric

Reason: For every (a,b) , (b,a) is also in R. Not antisymmetric because $1 \neq 3$ but both $(1,3)$ and $(3,1)$ exist.

Q5: If $x \equiv 17 \pmod{5}$, which integer is a valid solution for x ?

1. 12 ✓

2. -44

3. 8

4. 4

Reason: $17 \bmod 5 = 2$, and $12 \bmod 5 = 2$.

Q6: In matrix representation of a reflexive relation, the diagonal entries are always:

1. Transitive

2. Symmetric

3. 1 ✓ (Reflexive)

4. Null

Reason: Reflexive relations have (a,a) for all a , so diagonal = 1.

Q7: The function $f(x) = 1/[(x+2)(x-2)]$ from $\mathbb{Z}$ to $\mathbb{Z}$ is not well defined because:

1. Function is defined at $x=2$ but not defined at $x=-2$

2. Function gives imaginary values for $x < 0$

3. Each input has two outputs

4. Function is not defined at $x=-2$ and $x=2$ ✓

Reason: Denominator becomes 0 at $x = \pm 2$, so $f(x)$ is undefined there.

Q8: Let $f = \{(2, 6), (4, 2), (5, 1)\}$ from $X = \{2, 4, 5\}$ to $Y = \{1, 2, 4, 6\}$. The range of f is:

1. $\{1, 2, 4, 6\}$

2. $\{2, 4, 5\}$

3. $\{1, 4, 6\}$

4. $\{1, 2, 6\}$ ✓

Reason: Range is the set of second elements: 6, 2, 1.

Q9: The union of R and S is $R \cup S = \{(1,2), (1,3), (2,1), (2,2), (2,4), (3,3), (4,2), (4,3)\}$. Then $R \cup S$ is:

1. Not transitive ✓

2. Transitive

Reason: $(1,2)$ and $(2,4)$ are in $R \cup S$ but $(1,4)$ is not, so it fails transitivity.

Q10: Domain of a relation symbolically written as:

1. $\text{Dom}(R) = \{a \in A \mid (a,b) \notin R\}$
2. None of these
3. $\text{Dom}(R) = \{b \in B \mid (a,b) \in R\}$
4. **$\text{Dom}(R) = \{a \in A \mid (a,b) \in R\}$ ✓**